

Afnan Amin Ali

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WORK EXPERIENCE

Data Engineer | [Workhuman](#) | Dublin, Ireland

07/2023 - Present

- Designed and deployed a **Retrieval-Augmented Generation (RAG)** platform on **Amazon Bedrock**, **OpenSearch**, and enterprise knowledge bases; automated provisioning via reusable **Terraform modules** to enable scalable enterprise GenAI adoption.
- Architected a **Generic Processing Framework** in **Python** that consolidated new and legacy pipelines into a standardized model, transforming data from **raw to trusted zone** with quality checks, standardization, and enrichment logic.
- Expanded the **Generic Data Ingestion Framework (GDIF)** in **Scala on Amazon EMR** with reusable ingestion patterns, **MSSQL connector** support, and **incremental loading**, reducing compute consumption and improving data freshness across enterprise sources.
- Delivered a **global search data product** spanning **25+ years of historical data** with optimized partitioning and redesigned personalization jobs feeding **OpenSearch** indexes — improving query performance, search relevance, and cost efficiency.
- Built an end-to-end **real-time streaming PoC and MVP** using **Apache Flink**, **Flink CDC** (Oracle, PostgreSQL), **Kafka**, and **Apache Iceberg**, validating enterprise streaming strategy and informing production adoption.
- Developed a **reusable Flink framework** (job configs, sources/sinks, transformation patterns) that standardized streaming development and accelerated onboarding of new use cases.
- Modernized **DBT** for next-gen analytics on **Redshift Serverless** — Docker image changes, **DBT tagging**, and selective model execution via exclude parameters — improving governance and reducing unnecessary compute spend.
- Deployed and operated **Apache Airflow** in AWS data lake environments to orchestrate critical workflows, increasing automation and reducing manual intervention.
- Resolved recurring **DMS** and ingestion reliability incidents through retry mechanisms, timeout tuning, and root cause analysis across business-critical integrations (Salesforce, Customer.io, Microsoft 365, PostgreSQL, Oracle, MSSQL).
- Mentored new engineers via architecture walkthroughs and tooling guidance; participated in technical interviews to support hiring across the data platform org.

Data Engineer | [BazaarTech](#)

11/2021 – 02/2023

- Designed **federated query architecture using Trino**, enabling unified access across heterogeneous data sources.
- Built and scaled a 100+ TB distributed data platform supporting real-time analytics
- Added **Apache Pinot** to the platform to provide real-time analytics, allowing for very quick query responses for dashboards and reports on fast-moving event data.
- Worked on data security and governance using **Apache Ranger**.
- Developed dashboards and visualisations on **Apache Superset** to help business users analyse data.
- Designed SheetBots and Slackbots in PySpark for data updates every hour and orchestrated it using Airflow.

Software Engineer | [TimeXperts](#)

06/2021 – 11/2021

- Built AI/ML forecasting models for sales prediction and customer retention. Translated data insights into actionable business strategies, improving revenue outcomes.

- Developed high-availability C++ transaction systems compliant with global payment standards (VISA, Mastercard).
- Optimise performance and reliability in mission-critical financial systems.

TECHNOLOGIES AND LANGUAGES

Languages: Python, Scala, C++, Go

Data Engineering: Spark, Kafka, Iceberg, Airflow

Open Source Tools: Apache Flink, Apache Iceberg, Apache Kafka, Apache Hive, Apache Superset, Apache Hudi, Apache Airflow, DBT, Terraform, Grafana (Loki), OpenMetadata

MLOps & AI: AWS Bedrock, ML Pipelines, RAG, OpenSearch, Feature Engineering

Cloud & Infra: AWS, Kubernetes, Docker, Terraform

Databases: Trino, PostgreSQL, MySQL , Oracle

EDUCATION

NED University Of Engineering and Technology
BE, Computer System Engineering

2016–2020
3.72/4.0 CGPA